

Project Phase-II Report
On
Carbon Footprint Analyzer using AI

Submitted by

Vaishvani Gramopadhye

Prerna Dawada

Khushi Kasliwal

Under the guidance of

Mr. V. V. Anagire



Department of Electronics and Telecommunication Engineering

S.A. P.D. J. Pathashala's
Walchand Institute of Technology, Solapur
(An Autonomous Institute)
2025-26

CERTIFICATE

This is to certify that Project Phase I Report entitled

Carbon Footprint Analyzer using AI

Submitted by

Sr. No	Name	Exam Seat No.
31.	Gramopadhye Vaishnavi Nitin	2204111042
32.	Dawada Prerna Sachin	2204111050
33.	Kasliwal Khushi Manoj	2204111091

has been approved as the partial fulfillment for the award of degree in 'B.Tech Electronics and Telecommunication Engineering' at Walchand Institute of Technology, Solapur affiliated to Punyashlok Ahilyadevi Holkar Solapur University, Solapur in the academic year 2025-26.

Mr. V.V. Anagire
Project Guide

Dr. Ms. A.V. Thalange
Head, E&TC Engg.

Dr. V. A. Athavale
Principal

Department of Electronics and Telecommunication Engineering

S.A.P.D. J. Pathashala's
Walchand Institute of Technology, Solapur
(An Autonomous Institute)
2025-26

SPONSORSHIP CERTIFICATE



Newfab Industries

All Types Of Industrial Powder Coating

Plot No. H-34, M.I.D.C. Waluj, Aurangabad, - 431 136.
Mo. ~~9220044524~~, ~~9852300571~~, Email : bhagwatmg001@gmail.com

9657007010

Date : 01/12/2025

TO,

The Principal,
Walchand Institute of Technology,
Solapur – 413006

SUBJECT: TECHNICAL SPONSORSHIP FOR FINAL YEAR PROJECT

RESPECTED SIR,

With reference to the above subject, we are pleased to provide **technical sponsorship** for the final year project proposed by your students.

The project involves developing a **real-time AI-based Carbon Footprint Analyzer** using Machine Learning and Data Analytics to monitor and reduce carbon emissions. We will provide necessary **technical guidance and mentorship** throughout the project duration.

Following students are the members of the project team:

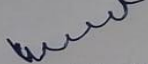
1. Vaishnavi Gramopadhye
2. Prerna Dawada
3. Khushi Kasliwal

Guide: Mr. V. V. Anagire

Industrial Guide: Mr. S. R. Kasliwal

Kindly approve and support the project.

Best Regards,


Santosh Kasliwal

Authorized Signatory

Newfab Industries





Newfab Industries

All Types Of Industrial Powder Coating

Plot No. H-34, M.I.D.C. Waluj, Aurangabad, - 431 136.
Mo. ~~98224171~~, ~~98224171~~ Email : bhagwatmg001@gmail.com

TECHNICAL SPONSORSHIP CERTIFICATE

This is to certify that **Newfab Industries, Aurangabad** has provided technical sponsorship and support for the final year engineering project titled "**Carbon Footprint Analyzer using AI**", carried out by the students of **Walchand Institute of Technology, Solapur**.

The project focuses on developing a real-time AI-based system for analyzing and predicting carbon emissions using parameters such as electricity consumption, transportation activities, and fuel usage. The organization has extended valuable technical guidance, mentorship, and support throughout the project duration.

The following students have successfully worked on this project:

1. Vaishnavi Gramopadhye
2. Prerna Dawada
3. Khushi Kasliwal

The project was completed under the academic guidance of **Mr. V. V. Anagire** and industrial guidance of **Mr. S. R. Kasliwal**.

We appreciate the efforts and commitment shown by the students and wish them success in their future endeavors.

Date: 02/05/2025

From,

Newfab Industries

Mr. S. R. Kasliwal



ACKNOWLEDGEMENT

We would like to express our sincere gratitude to all those who have supported and guided us in the successful completion of our project titled “**Carbon Footprint Analyzer using AI.**” This project has been a valuable learning experience, enhancing our understanding of Artificial Intelligence, Machine Learning, and web-based system development.

We are thankful to **Dr. V. A. Athavale**, Principal of Walchand Institute of Technology, Solapur, for providing us with the opportunity and necessary facilities to carry out this project.

We would also like to extend our gratitude to **Dr. A. V. Thalange**, Head of the Department of Electronics and Telecommunication Engineering, for her support and encouragement.

We are especially grateful to our project guide, **Mr. V. V. Anagire**, for his constant guidance, valuable suggestions, and encouragement throughout the project. His support was instrumental in completing this work successfully.

We also thank all faculty members and staff of the department for their cooperation.

Finally, we express our appreciation to our team members for their dedication, teamwork, and consistent efforts in completing this project.

ONE PAGE ABSTRACT

Carbon Footprint Analyzer using AI

Project Group : B5 G1

Guide :

Prof.V.V. Anagire

Name of Students :

1. Vaishnavi Gramopadhye
2. Prerna Dawada
3. Khushi Kasliwal

Sponsored By :

Newfab Industries

Plot No. H-34 M.I.D.C ,
Waluj, Aurangabad
431136.

Carbon emissions and climate change have become major global concerns, creating a need for efficient systems to monitor and manage carbon footprints. Increasing energy consumption, transportation activities, and fuel usage have significantly contributed to greenhouse gas emissions, making it essential to develop intelligent and data-driven solutions. The project “*Carbon Footprint Analyzer using AI*” presents a web-based application designed to analyze and predict carbon emissions based on user activity data.

The system allows users to upload datasets or manually input information related to key emission sources such as transportation, electricity consumption, and fuel usage. Using predefined emission factors, the application converts these inputs into carbon emissions measured in terms of carbon dioxide (CO₂) equivalents. This provides a standardized and accurate estimation of the user’s overall carbon footprint.

To enhance its functionality, the system integrates a Machine Learning model, specifically Linear Regression, to analyze historical data and predict future emission trends. This predictive capability enables users to understand how their emissions may change over time and supports proactive decision-making for reducing environmental impact. By incorporating AI, the system transforms from a simple calculator into an intelligent analytical tool. The backend of the system is developed using Python with the Flask framework, ensuring efficient data processing and seamless integration of machine learning models. The frontend is built using HTML, CSS, and JavaScript, providing an interactive and user-friendly interface. The application includes a dashboard that visually represents emission data through graphs and trends, allowing users to easily interpret their carbon footprint and identify major contributing factors. Additionally, the system promotes sustainable practices by generating a digital



INDEX

Chapter No.		Name of Chapter	Page No
		Abstract	1
Chapter 1		Introduction	2
	1.1	Introduction	2
	1.2	Purpose of the project	3
Chapter 2		Literature Survey	4
	2.1	Literature Survey	4
	2.2	Research Gap	6
	2.3	Problem Statement	7
	2.4	Objectives	7
	2.5	Scope of the project	7
	2.6	Sustainable Development Goals Addressed	8
Chapter 3		System Description	9
	3.1	System Description	9
	3.2	Methodology	12
	3.3	Complexity and Feasibility of Design	13
Chapter 4		System Implementation	14
	4.1	System Implementation	14
	4.2	Results and Discussions	16
	4.3	Challenges Faced and Overcome	19
	4.4	Impact of Project on Society and Environment	19
Chapter 5		Cost of Project	21
	5.1	Software Tools and Technology Stack	21
	5.2	Cost of Project	22

Chapter 6		Advantages, Disadvantages and Applications	23
Chapter 7		Conclusion and Future Scope	24
	7.1	Conclusion	24
	7.2	Future Scope	24
		Bibliography	25
		Appendix	26
		Paper published in conference or journal with certificate	28

List of Figures

Figure. No.	Title of Figure	Page No.
3.1.1	Block Diagram of Project	10
3.1.2	Flowchart of the Project	10
4.2.1	Home Page of the Project	16
4.2.2	Real Time Carbon Processing using API Data	17
4.2.3	Emission Analysis Dashboard	17
4.2.2	Generated Green Certificate	18

List of Tables

Table. No.	Title of Table	Page No.
5.1	Software Tools and Technology Stack	21
5.2	Cost of Project	22

ABSTRACT

Carbon emissions and climate change have become critical global challenges, creating a need for effective tools to monitor and manage carbon footprints. The project “Carbon Footprint Analyzer using AI” presents a web-based system designed to analyze and predict carbon emissions based on user activity data.

The system allows users to upload datasets or provide inputs related to transportation, electricity consumption, and fuel usage. Using predefined emission factors, the application calculates carbon emissions and applies a Machine Learning model, specifically Linear Regression, to predict future emission trends.

The backend of the system is developed using Python (Flask), while the frontend is built using HTML, CSS, and JavaScript, providing an interactive and user-friendly interface. The results are displayed through a dashboard, enabling users to understand their emission levels effectively.

Additionally, the system promotes sustainable practices by generating a digital certificate for users maintaining low carbon emissions. Although the current system operates on static input data, it is designed to be extended into a real-time automated system using APIs and advanced data integration techniques in the future.

This project demonstrates how Artificial Intelligence and data-driven approaches can contribute to environmental awareness and sustainable decision-making.

Chapter 1: Introduction

1.1: Introduction

In recent years, carbon emissions and climate change have emerged as critical global challenges, significantly impacting the environment and human life. The increasing use of fossil fuels, rapid urbanization, and industrial growth have contributed to a rise in greenhouse gas emissions, leading to global warming and environmental degradation. Monitoring and managing individual and organizational carbon footprints has therefore become essential for promoting sustainable development.

The “Carbon Footprint Analyzer using AI” project is designed to address this need by providing a smart, automated system that estimates carbon emissions based on user activities and environmental data. Unlike traditional methods that rely on static datasets and manual input, this system integrates real-time data and intelligent processing to deliver more dynamic and practical insights.

The system utilizes live environmental data, such as temperature, and combines it with activity-based estimations like transportation and energy consumption. Using predefined emission factors and intelligent logic, the system calculates the total carbon footprint and presents the results through an interactive web-based dashboard. Additionally, it provides meaningful insights and recommendations to help users reduce their carbon impact.

This project demonstrates the application of Artificial Intelligence and web technologies in solving real-world environmental problems. By transforming static analysis into a real-time, automated process, the system enhances usability, accuracy, and awareness, contributing toward a more sustainable future.

1.2: Purpose of Project

The primary objective of the “Carbon Footprint Analyzer using AI” is to develop an intelligent and automated platform that enables users to analyze, predict, and manage their carbon emissions effectively. The system is designed to provide meaningful insights into emission patterns and encourage environmentally sustainable practices.

The key objectives of the project are as follows:

- To analyze carbon emissions based on user-related and system-generated data such as transportation, electricity consumption, and fuel usage.
- To integrate real-time environmental data using APIs and process it dynamically for more accurate and practical analysis.
- To estimate and predict carbon emission trends using predefined emission factors and intelligent computational logic.
- To provide clear, interactive, and user-friendly insights through a web-based dashboard and visualizations.
- To encourage sustainable behavior by generating green certificates for users with low emission levels.
- To develop a scalable and automated system that supports continuous monitoring without manual data input.
- To create a foundation for future enhancements such as IoT integration, advanced Machine Learning models, and real-time decision support systems.

By achieving these objectives, the project promotes data-driven environmental awareness and contributes toward building a more sustainable and eco-friendly future.

Chapter 2: Literature Survey

2.1: Literature Survey

Carbon footprint assessment in educational institutions and buildings has been widely studied using various methodologies such as Life Cycle Assessment (LCA), emission factor analysis, and energy consumption models. Several research works and studies have contributed to understanding emission patterns and identifying major sources of carbon emissions.

1. A study on carbon footprint in an educational institution in India by R. Rahul et al. (IEEE) estimates approximately 2,129,736 kg CO₂ per year. It categorizes emissions into Scope 1, Scope 2, and Scope 3, identifying electricity consumption and mobile usage as major contributors. This study provided a foundation for understanding campus-level emission analysis.[1]
2. A case study on the carbon footprint of an academic building analyzes emissions from electricity, solar generation, diesel generators, and transportation fuel. The total emission is estimated to be around 559.9 metric tons of CO₂ per year, with electricity being the dominant source.[2]
3. A Life Cycle Assessment-based study conducted at BITS Pilani evaluates emissions across all scopes and highlights that indirect emissions, particularly from electricity and procurement, contribute significantly to the total carbon footprint.[3]
4. A study conducted at Clemson University analyzes campus emissions using an LCA approach and identifies electricity generation (~41%) and steam usage (~16%) as major contributors, emphasizing the importance of energy management.[4]A sustainability assessment at the
5. University of Technology, Jamaica (UTech) reports approximately 4,150.43 tCO₂e emissions, with around 94% attributed to electricity consumption, suggesting the need for renewable energy integration.[5]
6. A systematic review of carbon footprint studies in higher education institutions identifies electricity (78%), transportation (74%), and fuel usage (64%) as the most significant emission sources and proposes standardized evaluation procedures.[6]
7. Research on assessing carbon footprint in large universities provides methodologies for defining emission categories, data collection strategies, and assumptions to improve accuracy and comparability.[7]

8. A life cycle carbon emission study of a multi-purpose university building in Sri Lanka reveals that the operational phase contributes around 63% of emissions, while material production contributes approximately 31.6%, highlighting the importance of sustainable design.[8]
9. A study on carbon dioxide footprint in academic buildings uses sensor-based monitoring to analyze CO₂ concentration levels across different zones, showing higher emissions in office areas and enabling detailed footprint mapping.[9]
10. Research on building-level carbon footprint assessment evaluates emissions from construction, operation, and maintenance phases, recommending green building technologies such as solar energy and improved insulation to reduce long-term emissions.[10]

These studies primarily focus on static data analysis, Life Cycle Assessment techniques, and institutional-level carbon footprint evaluation. However, there is a lack of systems that provide real-time monitoring, automated data processing, and user-level insights. This gap highlights the need for intelligent and automated solutions, which is addressed in the proposed project.

2.2: Research Gap

Based on the review of existing literature on carbon footprint assessment in educational institutions and buildings, several limitations and research gaps have been identified.

Most of the existing studies focus on static carbon footprint analysis using historical or periodically collected data. These approaches provide a general overview of emissions but fail to offer continuous or real-time monitoring of carbon footprint.

Additionally, many studies rely on Life Cycle Assessment (LCA) methods, which, although accurate, are complex and not suitable for dynamic or user-level applications. These methods often require extensive data collection and are difficult to implement in real-time systems.

Another major limitation is the lack of automation in data collection and processing. Existing systems typically depend on manual data input or predefined datasets, which can lead to delays, inaccuracies, and reduced usability.

Furthermore, most research focuses on institutional-level analysis rather than individual or activity-based carbon tracking. There is limited emphasis on providing personalized insights or user-friendly interfaces for monitoring and reducing emissions.

There is also a noticeable absence of systems that integrate real-time environmental data with intelligent processing techniques. Current solutions do not effectively combine live data sources, computational models, and visualization tools into a single platform.

Moreover, existing systems often lack interactive dashboards and decision-support features that can help users understand their emission patterns and take corrective actions.

Therefore, there is a need for a smart, automated, and scalable system that can perform real-time carbon footprint analysis, reduce dependency on manual input, and provide meaningful insights through an interactive interface.

The proposed project addresses these gaps by developing a real-time, API-based carbon footprint analyzer that integrates automated data processing, intelligent estimation, and dynamic visualization, thereby enhancing both accuracy and usability.

2.3: Problem Statement

Develop real-time intelligent systems to monitor carbon emissions and enable predictive, automated compliance for timely sustainability efforts.

2.4: Objectives of the Project

The main objectives of the “Carbon Footprint Analyzer using AI” project are:

1. To develop a system that analyzes carbon emissions based on transportation, electricity consumption, and fuel usage.
2. To integrate real-time environmental data using APIs for dynamic and up-to-date analysis.
3. To eliminate manual data entry by implementing an automated data processing system.
4. To calculate carbon emissions using standard emission factors and intelligent computational logic.
5. To provide a user-friendly and interactive dashboard for visualizing carbon footprint data.

2.5: Scope of the project

The scope of the “Carbon Footprint Analyzer using AI” project focuses on developing a web-based system that enables automated and real-time estimation of carbon emissions based on environmental conditions and activity-based data.

The system is designed to analyze emissions from key sources such as transportation, electricity consumption, and fuel usage. It integrates live environmental data through APIs and combines it with computational logic to generate dynamic emission values without requiring manual input.

The project includes the development of an interactive dashboard that visualizes carbon emission data through charts, KPIs, and insights, making it easier for users to understand their environmental impact. It also provides a certification feature to encourage sustainable behavior based on emission levels.

The current scope is limited to simulated and estimated activity data combined with real-time environmental inputs, making it suitable for demonstration and analysis purposes. However, the system is designed to be scalable and can be extended in the future to include IoT-based real-time data collection, integration with smart devices, advanced Machine Learning models for prediction, and personalized recommendation systems.

Thus, the project establishes a foundation for a fully automated and intelligent carbon monitoring system that can be further expanded for real-world deployment.

2.6: Sustainable Development Goals Addressed

The “Carbon Footprint Analyzer using AI” project contributes to multiple United Nations Sustainable Development Goals (SDGs) by promoting environmental awareness, sustainable practices, and responsible resource management.

- **SDG 13: Climate Action**

This project directly addresses climate change by enabling users to monitor and analyze their carbon emissions. By providing real-time insights and encouraging emission reduction, it supports efforts to combat global warming.

- **SDG 7: Affordable and Clean Energy**

The system highlights the impact of electricity consumption on carbon emissions, promoting awareness about efficient energy usage and encouraging the adoption of cleaner and renewable energy sources.

- **SDG 12: Responsible Consumption and Production**

By analyzing emissions from daily activities such as transportation and fuel usage, the system encourages users to make more responsible and sustainable consumption choices.

- **SDG 9: Industry, Innovation and Infrastructure**

The project leverages modern technologies such as Artificial Intelligence, APIs, and web-based platforms to develop an innovative and scalable solution for environmental monitoring.

- **SDG 11: Sustainable Cities and Communities**

By promoting reduced emissions and sustainable behavior, the system contributes to creating environmentally friendly and sustainable urban communities.

Chapter 3: System Description

3.1: System Description

The proposed Carbon Footprint Analyzer using AI is a structured and automated system designed to estimate carbon emissions using real-time environmental data and intelligent computational processing. The system architecture is divided into multiple interconnected layers, ensuring smooth data flow from collection to visualization and reporting.

The system begins with the Data Sources Layer, which includes real-time environmental data obtained through APIs such as weather data (temperature, humidity, and conditions), along with internally generated activity data such as transportation distance, fuel usage, and electricity consumption. These inputs form the foundation for further analysis.

The collected data is passed to the Data Collection Layer, where real-time environmental parameters and simulated activity data are organized into structured formats. This layer ensures that both external and internal data sources are properly integrated into the system.

Next, the data is processed in the Processing Engine, which performs data cleaning and validation to ensure accuracy. It then transforms raw data into meaningful parameters. For example, temperature values are used to estimate electricity consumption, and travel data is used to calculate fuel usage. This step converts raw inputs into usable features for emission calculation.

The processed data is then sent to the Carbon Calculation Module, where standard emission factors are applied. The system calculates carbon emissions from different sources such as transportation, electricity, and fuel consumption. These values are combined to determine the total carbon footprint of the user or system.

The calculated results are stored in the Data Storage Layer, where weekly records are maintained for future analysis and tracking. This allows the system to maintain historical data and support trend analysis.

The output is displayed through the Visualization and User Interface Layer, which presents the results in an interactive dashboard. This includes key performance indicators (KPIs), charts, graphs, and real-time insights that help users understand their carbon emission patterns.

Finally, the system includes a Certificate Module, which generates a sustainability certificate based on emission levels. This feature encourages users to adopt environmentally friendly practices by recognizing low emission performance.

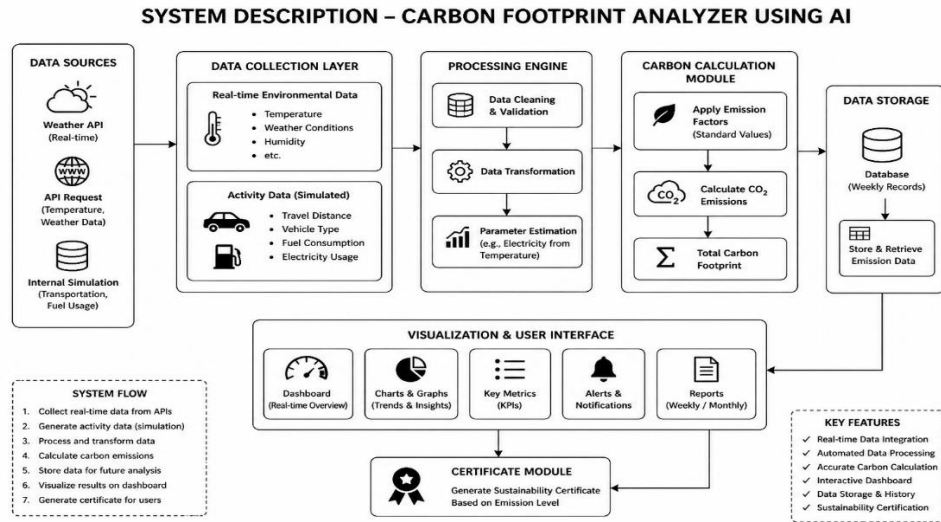


Fig 3.1.1: Block Diagram of the project

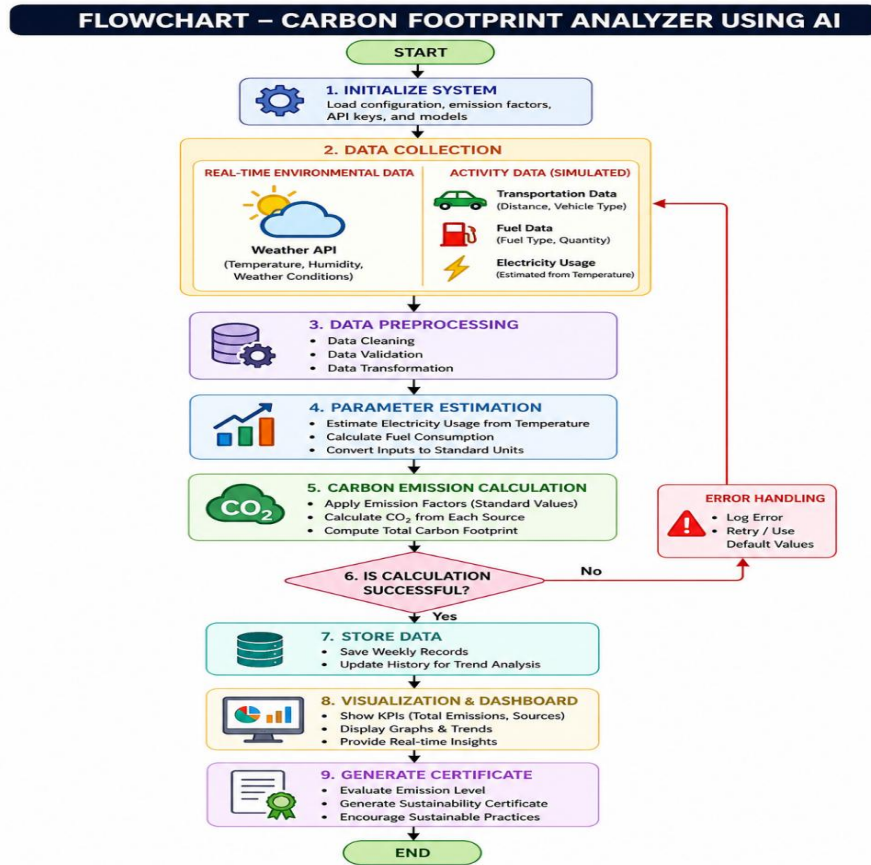


Fig 3.1.2: Flowchart of the project

Workflow:

The flowchart represents the working process of the Carbon Footprint Analyzer using AI, showing how data flows through different stages of the system from initialization to final output.

The process begins with the Start node, where the system is initialized by loading required configurations such as emission factors, API keys, and processing logic.

In the Data Collection stage, the system gathers two types of data. First, real-time environmental data such as temperature, humidity, and weather conditions is fetched using APIs. Second, activity-based data such as transportation distance, fuel usage, and electricity consumption is generated through internal simulation logic.

The collected data then moves to the Data Preprocessing stage, where it is cleaned, validated, and transformed into a structured format suitable for further analysis.

Next, in the Parameter Estimation stage, the system derives meaningful values from raw inputs. For example, electricity usage is estimated based on temperature, and fuel consumption is calculated from transportation data.

The processed data is then passed to the Carbon Emission Calculation stage, where standard emission factors are applied to compute carbon emissions from each source. These values are combined to determine the total carbon footprint.

A decision point checks whether the calculation is successful. If an error occurs, the system performs Error Handling, where it logs the error and either retries the process or uses default values to ensure continuity.

If the calculation is successful, the system proceeds to the Data Storage stage, where emission data is stored for weekly tracking and historical analysis.

The results are then displayed in the Visualization and Dashboard stage, where users can view key metrics, charts, and real-time insights about their carbon emissions.

Finally, the system reaches the Certificate Generation stage, where a sustainability certificate is generated based on emission levels to encourage environmentally friendly behavior.

The process ends with the End node, completing the cycle of real-time carbon footprint analysis.

Overall, the flowchart clearly illustrates the automated, real-time, and structured workflow of the system.

3.2: Methodology

The methodology of the Carbon Footprint Analyzer using AI project follows a systematic approach that includes data collection, processing, calculation, and visualization to estimate carbon emissions in real time.

Data Collection

The system collects real-time environmental data such as temperature using APIs. In addition, activity-based data such as transportation distance, fuel usage, and electricity consumption is generated using internal simulation logic. This ensures that the system operates without manual input.

Data Preprocessing

The collected data is cleaned, validated, and transformed into a structured format. This step ensures that the data is accurate and suitable for further analysis.

Parameter Estimation

The system derives meaningful parameters from raw data. For example, electricity consumption is estimated based on temperature, and fuel consumption is calculated using travel distance and efficiency factors.

Carbon Emission Calculation

The system applies standard emission factors to calculate carbon emissions from different sources such as transportation, electricity, and fuel usage. These values are combined to determine the total carbon footprint.

Error Handling

The system checks for errors during processing and calculation. In case of failure, it logs the error and either retries the process or uses default values to ensure system stability.

Data Storage

The calculated emission data is stored periodically (weekly) for future analysis and tracking of emission trends.

Visualization and Dashboard

The processed data is displayed through an interactive dashboard using charts, graphs, and key performance indicators (KPIs), enabling users to understand their carbon footprint easily.

Certificate Generation

Based on emission levels, the system generates a sustainability certificate to encourage users to adopt eco-friendly practices.

3.3: Complexity and Feasibility of Design

The design of the Carbon Footprint Analyzer using AI is moderately complex, as it involves the integration of multiple components such as real-time APIs, data processing modules, emission calculation logic, and interactive visualization. However, the system is structured in a modular manner, which simplifies development, testing, and maintenance.

From a computational perspective, the system does not require heavy processing or high-end hardware. The calculations are based on standard emission factors and basic mathematical operations, making the system efficient and lightweight. The use of APIs for data collection ensures real-time functionality without increasing system complexity significantly.

The implementation complexity is manageable, as the system is developed using widely available technologies such as Python, Flask, and JavaScript. These technologies provide strong support for web development, API integration, and data visualization, reducing development effort.

In terms of feasibility, the project is highly practical and implementable. The system requires minimal hardware resources and can run on standard computing devices. The use of open-source tools and publicly available APIs further enhances its feasibility by reducing cost and dependency on proprietary systems.

The design is also scalable, allowing future enhancements such as integration with IoT devices, advanced machine learning models, and real-time monitoring systems. This makes the system adaptable for both academic and real-world applications.

Overall, the proposed design achieves a balance between functionality and simplicity, making it both technically feasible and efficient for real-time carbon footprint analysis.

Chapter 4 : System Implementation

4.1: System Implementation

The implementation phase of the Carbon Footprint Analyzer using AI focuses on transforming the designed architecture into a fully functional, real-time web-based system. The system integrates backend processing, API-based data acquisition, frontend visualization, and automated features to provide dynamic carbon emission analysis.

4.1.1 Backend Implementation

The backend of the system is developed using Python with the Flask framework. It acts as the core processing unit and is responsible for handling requests, processing data, and generating responses.

The system fetches real-time environmental data using APIs.

It processes activity-based data such as transportation distance and fuel usage using internal logic.

Carbon emission calculations are performed using predefined emission factors.

Flask routes such as /realtime, /store_weekly, and /certificate are implemented for data handling and functionality execution.

4.1.2 Data Handling and Storage

Instead of using a heavy database, the system uses lightweight in-memory or file-based storage for managing weekly emission data.

Emission data is stored periodically for tracking and analysis.

This approach keeps the system simple, fast, and efficient for real-time applications.

4.1.3 Data Processing and Simulation Module

The system includes a processing engine that transforms raw inputs into meaningful parameters.

Temperature data is used to estimate electricity consumption.

Transportation distance is used to calculate fuel usage.

This module eliminates the need for manual input and ensures continuous automated operation.

4.1.4 Carbon Emission Calculation Module

The system calculates emissions using standard emission factors.

Emissions from transportation, electricity, and fuel consumption are calculated separately.

These values are combined to determine the total carbon footprint in real time.

4.1.5 Frontend and Visualization Implementation

The frontend is developed using HTML, CSS, and JavaScript to provide an interactive user interface.

Real-time data is fetched from the backend using asynchronous JavaScript (fetch API).

The dashboard displays KPIs, emission values, and insights dynamically.

Charts such as pie charts and line graphs are used to visualize emission distribution and trends.

4.1.6 Certificate Generation Module

The system includes a certificate generation feature implemented using the ReportLab library in Python.

A PDF certificate is generated based on emission levels.

This feature encourages users to adopt sustainable practices.

4.1.7 Integration and Testing

All modules including backend, processing engine, visualization, and certificate generation are integrated into a unified system.

Functional testing ensures correct emission calculations.

Integration testing verifies smooth communication between frontend and backend.

System testing ensures stable real-time performance without errors.

4.1.8 Deployment

The system is deployed locally using a Flask server.

The backend runs on a local host environment.

The web interface is accessed through a browser.

The system can be extended for cloud deployment in the future.

Overall, the implementation ensures a real-time, automated, and user-friendly system that effectively analyzes and visualizes carbon emissions without manual intervention.

4.2: Results and Discussions

The Carbon Footprint Analyzer using AI was successfully implemented as a real-time web-based system. The system is able to fetch environmental data through APIs, process it automatically, and calculate carbon emissions without manual input.

The dashboard displays real-time results using KPIs, charts, and graphs, making it easy to understand emission patterns. The system updates data periodically and provides dynamic visualization of transportation, electricity, and fuel-related emissions.

The certificate generation feature was also successfully implemented, allowing users to download a sustainability certificate based on their emission levels.

The results show that the system works efficiently and provides accurate and user-friendly insights. However, since some data is simulated, further improvements can be made by integrating real-world data sources.

Overall, the system effectively demonstrates real-time carbon footprint analysis and visualization.

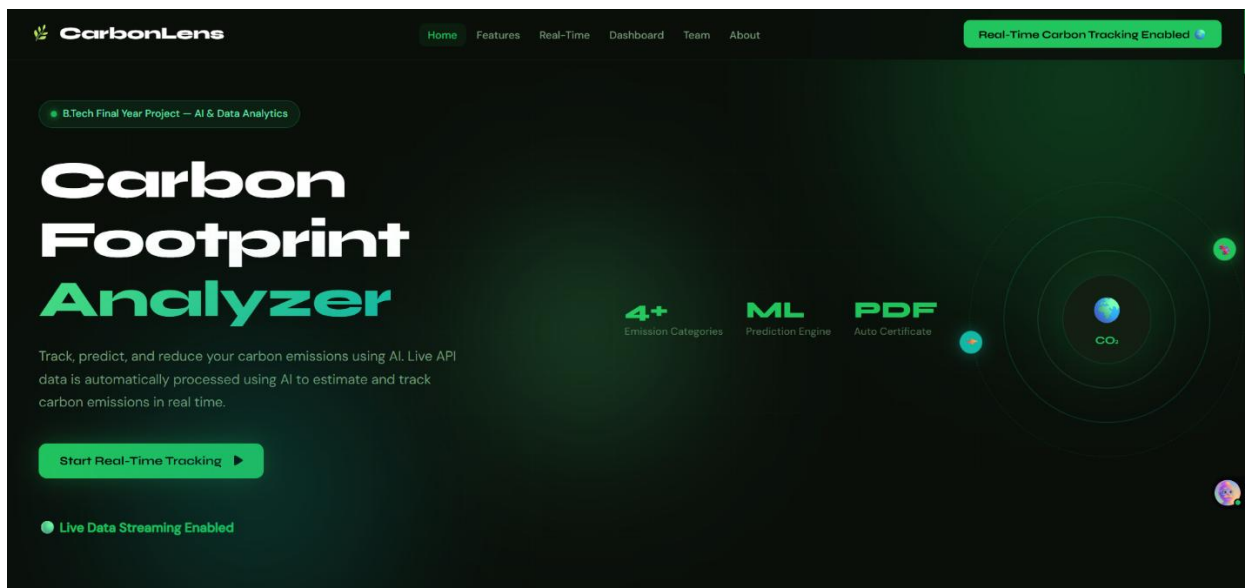


Fig. 4.2.1: Home Page of Carbon Footprint Analyzer

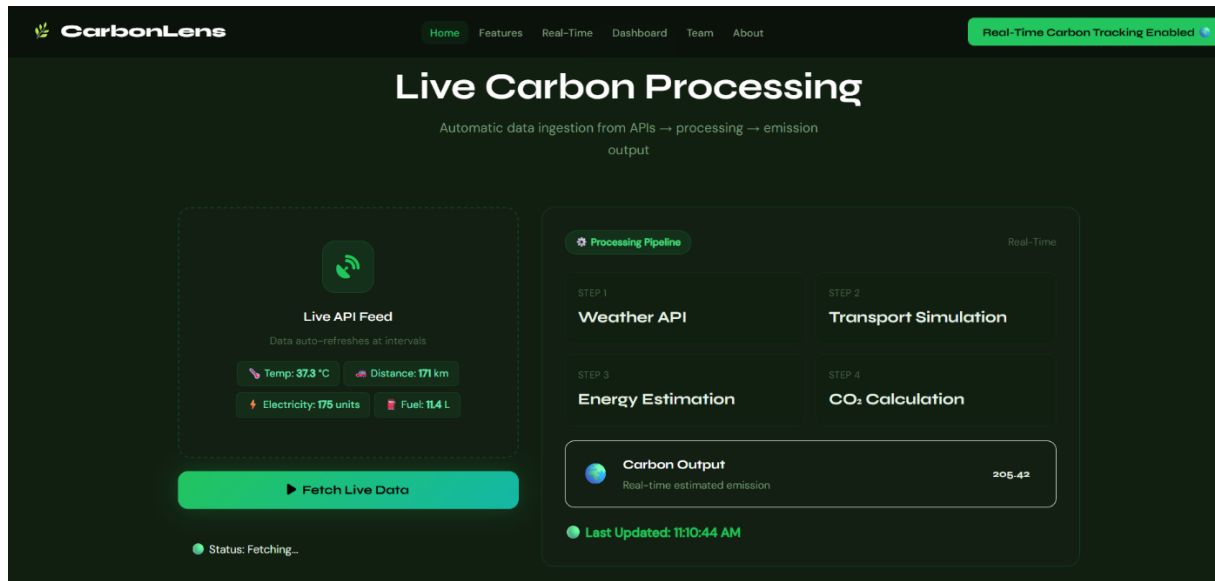


Fig. 4.2.2: Real-Time Carbon Processing using API Data

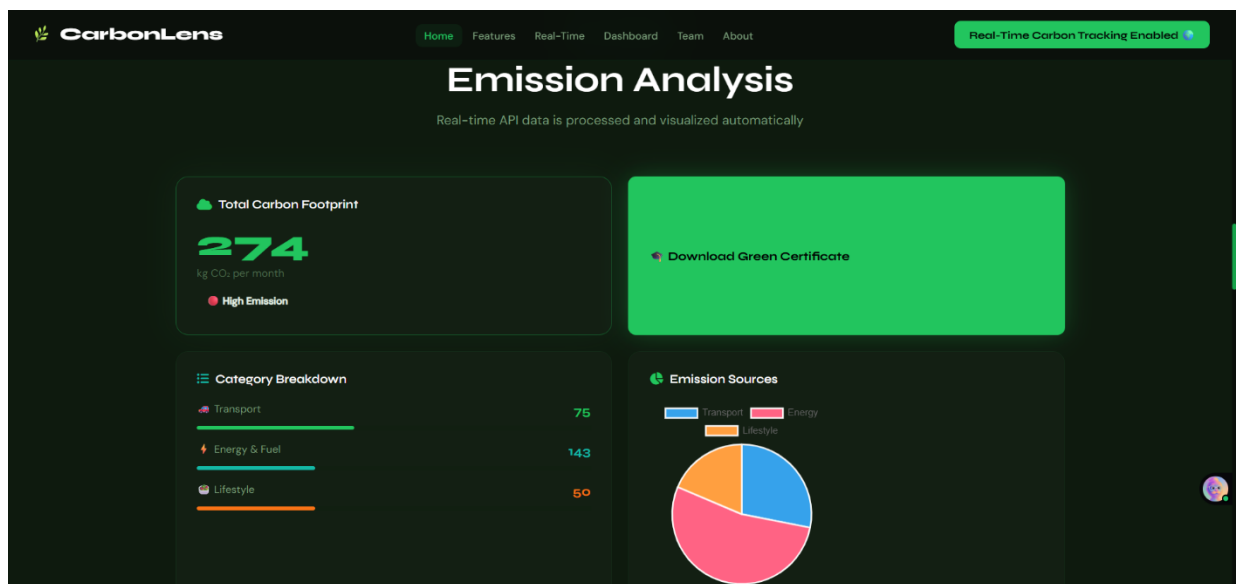


Fig. 4.2.3: Emission Analysis Dashboard



Fig. 4.2.4: Generated Green Certificate

4.3: Challenges Faced and Solutions

- **Challenges Faced**

- a. Difficulty in integrating real-time API data and handling continuous data updates
- b. Converting raw environmental data into meaningful parameters (electricity, fuel usage)
- c. Ensuring smooth communication between frontend and backend in real time
- d. Handling errors during API failure or data fetching issues
- e. Designing a clear and interactive dashboard for data visualization
- f. Implementing dynamic certificate generation based on emission values

- **Solutions Implemented**

- a. Implemented API integration with proper request handling and periodic data fetching
- b. Developed estimation logic to convert temperature and distance into usable parameters
- c. Used asynchronous JavaScript (fetch API) to enable real-time frontend updates
- d. Added error handling mechanisms such as retry logic and default fallback values
- e. Designed a structured UI with charts, KPIs, and real-time insights for better understanding
- f. Used ReportLab library in Python to generate automated PDF sustainability certificates

4.4: Impact of Project on Society and Environment

- **Impact on Society**

The Carbon Footprint Analyzer using AI has a positive impact on society by increasing awareness about carbon emissions and promoting sustainable behavior. It provides users with a simple and interactive platform to understand their environmental impact in real time.

The system encourages individuals to make better lifestyle choices, such as reducing fuel consumption and optimizing electricity usage. By visualizing emissions through dashboards and insights, users can easily identify major sources of pollution and take corrective actions.

The inclusion of a sustainability certificate further motivates users by recognizing eco-friendly behavior. Overall, the project contributes to building environmental awareness and supports responsible decision-making in society.

- **Impact on Environment**

The project has a direct positive impact on the environment by promoting the reduction of carbon emissions. By identifying major emission sources such as transportation and energy consumption, the system helps users take steps to minimize their carbon footprint.

Since the system is software-based, it requires minimal physical resources and operates on standard computing devices. It reduces dependency on manual processes and supports a paperless approach, contributing to eco-friendly practices.

Additionally, the system is energy-efficient and scalable, making it suitable for wider applications such as organizations and smart cities. Overall, the project supports environmental sustainability and contributes toward a greener future.

Chapter 5 : Cost of Project

5.1: Software Tools and Technology Stack

Category	Technology /Tool	Purpose
Programming Language	Python	Backend development, data processing, emission calculations
Web Framework	Flask	Creating server, handling requests, API endpoints
Frontend	HTML, CSS, JavaScript	Designing UI and enabling real-time interaction
API Integration	OpenWeather API	Fetching real-time environmental data
Data Visualization	Chart.js / PowerBI	Creating charts and graphs for dashboard
PDF Generation	ReportLab	Generating sustainability certificate in PDF format
Development Tool	VS Code	Writing and managing code
Testing Platform	Web Browser (Chrome/Edge)	Running and testing the application
Version Control	Git/GitHub	Managing project versions and collaboration

5.2: Cost of Project

Resource Type	Particulars / Tools	Estimated Cost (₹)	License Type	Remarks	Purpose
Hardware Resources	Laptop / PC (Intel i5, 4 GB RAM, 256 GB SSD), Internet Connection	₹46,500	—	Used for development, testing, and running the application	Development & Execution
Software Resources	Python, Flask, VS Code	₹0	Open Source	Freely available tools used for backend development	Backend Development
Frontend Technologies	HTML, CSS, JavaScript	₹0	Open Source	Used to design responsive and interactive user interface	UI Design & Interaction
API Integration	OpenWeather API	₹0 (Free Tier)	Freemium	Used to fetch real-time environmental data	Real-Time Data Collection
Data Visualization	Chart.js	₹0	Open Source	Used for creating charts and dashboards	Data Visualization
PDF Generation	ReportLab	₹0	Open Source	Used to generate sustainability	Certificate Generation
Storage / Data Handling	File-based storage (JSON / temporary storage)	₹0	Open Source	Used for storing weekly emission data	Data Storage
Development Effort	Design, Coding, Testing, Integration	₹3,000 (Approx.)	—	Estimated effort cost for project implementation	System Development
Deployment	Localhost / Basic Hosting	₹0	—	Project deployed locally for demonstration	Execution Environment
Total Estimated Cost	Including hardware	₹49,500	—	Project completion cost	—
Estimated Software Cost	—	₹3,000 (Approx.)	—	Project completion cost	—

Chapter 6 : Advantages , Disadvantages and Applications

6.1 Advantages

- Provides real-time carbon emission analysis using live API data
- Eliminates the need for manual data entry through automation
- User-friendly interface with interactive dashboard and visualizations
- Helps users understand and reduce their carbon footprint
- Lightweight and cost-effective system using open-source technologies
- Encourages sustainable behavior through certificate generation
- Scalable design that can be extended for advanced applications

6.2 Disadvantages

- Some activity data is simulated, which may affect accuracy
- Depends on internet connectivity for API data fetching
- Limited to estimation-based calculations rather than exact measurements
- Does not include direct integration with IoT or real-world sensors
- Basic AI logic; lacks advanced machine learning-based predictions

6.3 Applications

- Personal carbon footprint monitoring for individuals
- Educational institutions for awareness and analysis of emissions
- Organizations for basic sustainability tracking
- Smart city initiatives for environmental monitoring (future scope)
- Research and academic projects related to climate change
- Awareness platforms to promote eco-friendly practices

Chapter 7: Conclusion and Future Scope

7.1 Conclusion

The Carbon Footprint Analyzer using AI project successfully demonstrates a real-time, automated approach for analyzing carbon emissions. The system overcomes the limitations of traditional static methods by integrating live API data, intelligent processing, and interactive visualization. It provides users with meaningful insights into their carbon footprint through a user-friendly dashboard and graphical representations.

The implementation of automated data handling and real-time updates makes the system efficient and practical for continuous monitoring. The certificate generation feature further encourages users to adopt sustainable practices. Overall, the project highlights how technology can be effectively used to promote environmental awareness and responsible decision-making.

7.2 Future Scope

- Integration with IoT devices and smart sensors for accurate real-world data collection
- Use of advanced Machine Learning models for emission prediction and pattern analysis
- Integration with GPS and maps for real-time transportation tracking
- Development of mobile application for wider accessibility
- Personalized recommendations based on user behavior and emission trends
- Deployment on cloud platforms for large-scale usage
- Integration with smart city systems and organizational monitoring platforms

Bibliography

- [1] R. Rahul *et al.*, “Carbon Footprint of an Academic Building – A Case Study,” *IJERT*, vol. 10, no. 7, 2021. [Online].
Available: <https://www.ijert.org/carbon-footprint-of-an-academic-building-a-case-study>
- [2] G. Hammond and C. Jones, “Embodied Energy and Carbon in Construction Materials,” *Energy Procedia*, vol. 134, 2017. [Online].
Available: <https://www.sciencedirect.com/science/article/pii/S2212827117308910>
- [3] P. Choudhary *et al.*, “Assessment of Carbon Footprint in Educational Institutes,” *ResearchGate*, 2020. [Online].
Available: <https://www.researchgate.net/publication/342608206>
- [4] M. M. Carballo and J. D. C. Solís, “Framework for Carbon Footprint Estimation in University Campuses,” *LACCEI Conf.*, Costa Rica, 2024. [Online].
Available: https://laccei.org/LACCEI2024-CostaRica/papers/Contribution_1882_final_a.pdf
- [5] ResearchGate, “Carbon Footprint and Sustainability Studies.” [Online].
Available: <https://www.researchgate.net>
- [6] M. Caldera *et al.*, “Energy and Environmental Management Systems for Carbon Reduction,” *Energies*, vol. 16, no. 1, 2023. [Online].
Available: <https://cris.unibo.it/retrieve/.../energies-16-00166-v2.pdf>
- [7] X. Li *et al.*, “Carbon Emission Assessment and Mitigation in Buildings,” *Front. Eng. Manag.*, vol. 5, no. 4, 2018. [Online].
Available: <https://journal.hep.com.cn/fem/EN/10.15302/J-FEM-2018055>
- [8] R. R. Tan *et al.*, “Optimization for Carbon Footprint Reduction,” *Sustainability*, vol. 13, no. 14, p. 7847, 2021. [Online].
Available: <https://www.mdpi.com/2071-1050/13/14/7847>
- [9] A. Verma and S. Sharma, “Carbon Footprint Analysis and its Impact on Climate Change,” *IRJMETs*, vol. 5, no. 3, 2023. [Online].
Available: https://www.irjmets.com/uploadedfiles/paper/.../fin_irjmets1678281215.pdf

APPENDIX

The appendix section provides additional supporting details related to the implementation, working, and outputs of the Carbon Footprint Analyzer using AI project.

Appendix A: System Screenshots

This section includes visual representations of the developed system:

- Home Page (CarbonLens Interface): Displays the project title, navigation bar, and entry point to the system.
- Real-Time Data Processing Section: Shows live API-based data such as temperature, transportation distance, fuel usage, and electricity estimation.
- Dashboard Visualization: Displays key performance indicators (KPIs), emission values, pie charts, and trend graphs for analysis.
- Certificate Generation Output: Shows the generated sustainability certificate based on emission levels.

These screenshots demonstrate the working and user interface of the system.

Appendix B: Sample Input and Output

This section includes sample data used by the system:

• Input Data:

Temperature (from API)

Transportation distance (simulated)

Fuel usage (calculated)

Electricity consumption (estimated)

• Output Data:

Total carbon emission (kg CO₂)

Category-wise emissions (transport, energy, lifestyle)

Dashboard charts and insights

Example:

Temperature = 35°C

Distance = 200 km

→ Calculated Emission $\approx$ 220–270 kg CO₂

Appendix C: Key Code Snippets

This section includes important parts of the implementation:

- API Integration Code: Fetching real-time weather data using OpenWeather API
- Carbon Calculation Logic: Formula used to compute emissions from different sources
- Frontend Integration: JavaScript fetch API used for real-time updates
- Certificate Generation Code: Python ReportLab code used to generate PDF certificates

(Note: Complete source code is available in the project files.)

Appendix D: System Configuration

The system was developed and tested with the following configuration:

- Operating System: Windows 10 / 11
- Programming Language: Python 3.x
- Framework: Flask
- Frontend: HTML, CSS, JavaScript
- Libraries: Chart.js, ReportLab
- API Used: OpenWeather API
- Browser: Google Chrome / Microsoft Edge

Appendix E: Project Workflow Summary

The system follows this workflow:

Data Collection → Data Processing → Emission Calculation → Dashboard Visualization →
Data Storage → Certificate Generation

This workflow ensures real-time and automated analysis of carbon emissions.

Overall, the appendix provides additional clarity and technical support to understand the complete working of the system.

